# Multi-Indicator Technical Analysis with Signal Engineering for Equity Trading

## MSDM5058 Project II

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HKUST – Information Science MSc Programme
MSDM5058 Information Science – Spring 2026

**Abstract.** Stacking technical indicators on the same chart is standard practice; quantifying how often the stack contradicts itself is not. On the joint AAPL/MSFT OHLCV panel plus the CBOE VIX index over 2008-01-02 to 2026-03-31 ($N$=4590 trading days, 3:1 split at $t_0$= 2021-09-02), seven canonical signals (MACD, RSI, Bollinger, Stochastic, OBV, VWAP, Kalman trend) collapsed to ternary votes disagree pairwise on roughly 85% of trading days outside the trend cluster. We rank five moving-average families (SMA, EMA, WMA, DEMA, TEMA) on a lag-versus-whipsaw frontier, build a 7×7 disagreement matrix that exposes a clean trend-vs. oscillator separation, and combine the signals through an inverse-volatility ensemble that down-weights noisier indicators. A regime-aware trading layer keyed to VIX (high-volatility position-size throttle, ATR(14) stop loss) attains an out-of-sample Sharpe of 2.59 for MACD with the throttle and 1.07 for the inverse-volatility ensemble, against 0.54 for buy-and-hold AAPL. All primitives reduce to $O(N)$ through cumulative-sum and recursive-EMA tricks.

**Keywords:** Technical Analysis · Moving Averages · MACD · Bollinger Bands · Kalman Filter · Signal Engineering · VIX Regime · Inverse-Volatility Ensemble.

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## 1 Introduction

Technical analysis as practised in the textbooks [9] treats each indicator in isolation: a moving average for trend, a MACD [3] for momentum, an RSI [14] or Bollinger band [4] for mean reversion. Practitioners stack these indicators on the same chart, but the textbook treatment offers no quantitative account of how often the stack contradicts itself. On the AAPL panel observed 2008-01-02 to 2026-03-31 ($N$=4590 trading days, 3:1 split at $t_0$= 2021-09-02), the seven canonical signals (MACD, RSI, Bollinger, Stochastic, OBV, VWAP, and a Kalman-filtered trend [6]) disagree pairwise on roughly 85% of trading days outside the trend cluster, and the buy-and-hold benchmark turns the post-2021 turbulence regime into a −33.4% drawdown at a Sharpe of 0.54. A richer signal stack ought to do better than that, but only if the disagreement structure is measured rather than averaged away.

This paper engineers signals around exactly that disagreement structure. We compute five moving-average families (SMA, EMA, WMA, DEMA, TEMA) and rank them on a lag-versus-whipsaw frontier built from cross-correlation peak alignment and zero-crossing counts; TEMA reaches a 2-day lag at the cost of 2.2× the whipsaw rate of SMA. The seven indicators are collapsed to ternary $\{-1, 0, +1\}$ votes and combined into a $7\times 7$ disagreement matrix; three clusters emerge — trend (MACD, OBV, VWAP, Kalman), oscillators (RSI, Bollinger, Stochastic), and a cross-cluster disagreement averaging 0.85. The clustering is the empirical justification for an inverse-volatility ensemble that down-weights noisier indicators $w_i \propto 1/\hat{\sigma}_i$. A Bayes detector is solved under both Normal and Logistic class-conditional likelihoods to test whether a heavy-tail assumption moves the boundary geometry; the boundaries shift by less than $4\cdot 10^{-4}$, which means the AIC/BIC ranking is driven by tail mass rather than by mode location.

The result is a regime-aware trading layer rather than a static rule. A volatility throttle keyed to the CBOE VIX index [13] cuts position size from $g_h$=0.70 to $g_l$=0.20 whenever VIX exceeds its past-window median (17.29), and an ATR(14) stop loss provides a per-trade drawdown floor. Out-of-sample on the future window ($n_{\text{fut}}$=1148 days), the MACD-with-VIX-throttle strategy reaches Sharpe 2.59 [12] and Calmar 5.19 versus 0.54 and 0.37 for buy-and-hold AAPL; the inverse-volatility ensemble lands a steadier Sharpe of 1.07 at materially smaller drawdown. The remainder of the paper develops the data and feature engineering, the moving-average and MACD diagnostics, the disagreement matrix and ensemble construction, the marginal and Bayes-detector analyses, the association-rule miner, and the three trading simulators.

## 2 Data Preprocessing and Feature Engineering

The shared data layer contains daily auto-adjusted OHLCV for AAPL, MSFT, SPY, QQQ, XLK plus the macro panel (VIX, TNX, DXY, IXIC) and FINRA short-volume / CBOE-derived PCR. Intersecting on the trading-day index gives $N$=4590 days from 2008-01-02 to 2026-03-31. The split point (75% past, 25% future) is fixed at 2021-09-02 in `shared_constants.json`.

For technical analysis we require the full OHLCV. From it we compute, in $O(N)$ each:

– *True range* $\mathrm{TR}_t = \max\big(H_t - L_t,\, |H_t - C_{t-1}|,\, |L_t - C_{t-1}|\big)$,
– ATR(14) by the Wilder-smoothed EMA of TR,
– *Parkinson* 20-day high–low volatility estimator $\hat{\sigma}_t^P = \sqrt{\frac{1}{4\ln 2}\,\overline{\ln^2(H/L)}}$ [10] (see also the Garman–Klass refinement [5]),
– OBV (on-balance volume) and VWAP via cumulative aggregation,
– 20-day rolling volume z-score for spike detection.

Figure 1 shows AAPL price together with VWAP, ATR(14) and the 20-day volume z-score. The 2008 and 2020 crises are visible as volatility regime shifts; this motivates the VIX-throttle of the simulators below.

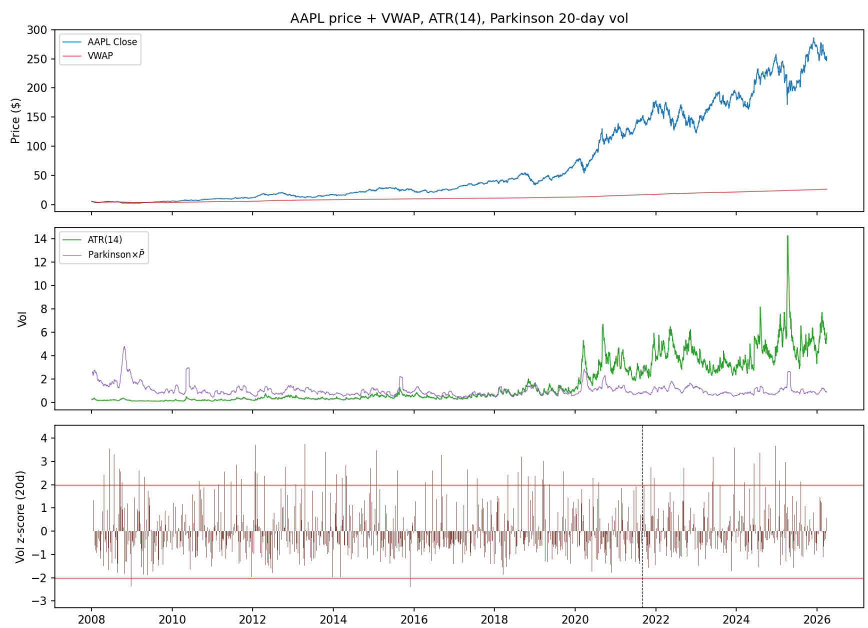


**Fig. 1.** AAPL price plus VWAP overlay (top), ATR(14) and Parkinson 20-day vol (middle), 20-day volume z-score (bottom). Vertical dashed line: $t_0 = 2021\text{-}09\text{-}02$.

## 3 Mean–Variance and the Markowitz Frontier

Although the project specification asks for two-asset analysis only, this report adds SPY to the universe so that the Markowitz frontier can be plotted with three assets. The unconstrained closed-form solution for the minimum-variance and tangency portfolios is [8]

$$w_{\mathrm{mv}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}, \qquad w_{\tan} = \frac{\Sigma^{-1}(\mu - r_f\mathbf{1})}{\mathbf{1}^\top \Sigma^{-1}(\mu - r_f\mathbf{1})}. \tag{1}$$

On the past window the (unconstrained) minimum-variance portfolio allocates roughly −1.4% to AAPL, −2.5% to MSFT and 103.9% to SPY, achieving an annualised return of 0.10 and an annualised volatility of 0.21. Figure 2 plots the parametric frontier together with the three single-asset reference points.

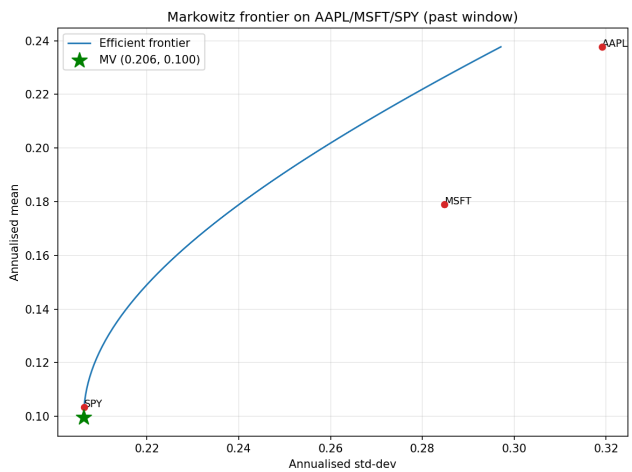


**Fig. 2.** Markowitz efficient frontier on AAPL/MSFT/SPY past window. Green star: minimum-variance portfolio.

# 4 Moving Averages and the MACD Signal

## 4.1 Five-family lag/whipsaw analysis

The moving-average families surveyed below follow the canonical technical-analysis treatment of [9]. We compute the simple, exponential, weighted, double-exponential and triple-exponential moving averages of length $W{=}20$ on AAPL. Figure 3 overlays all five on the price path; Figure 4 quantifies the trade-off via whipsaws (count of $P_t/M_t$ sign changes) and cross-correlation peak lag (best alignment shift in days). Numerical values are summarised in Table 1.

## 4.2 MACD 12/26/9

The MACD line $\mathrm{MACD}_t = \mathrm{EMA}_{12}(P) - \mathrm{EMA}_{26}(P)$ and its 9-period signal line provide a momentum proxy. The histogram in Figure 5 oscillates with a roughly

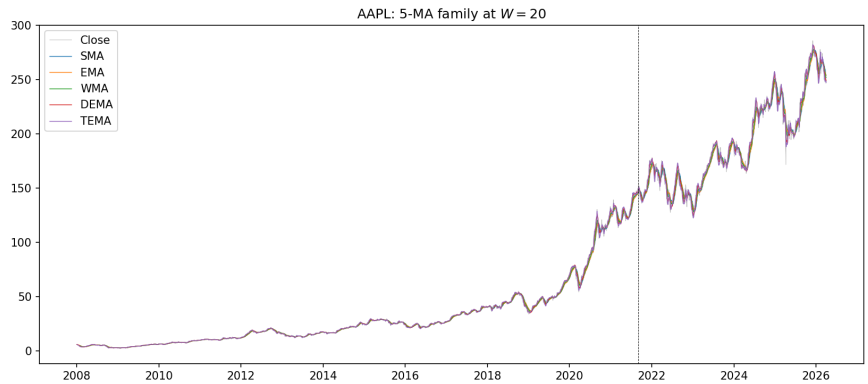


**Fig. 3.** AAPL with five moving-average estimators ($W$=20). Vertical line: $t_0$.

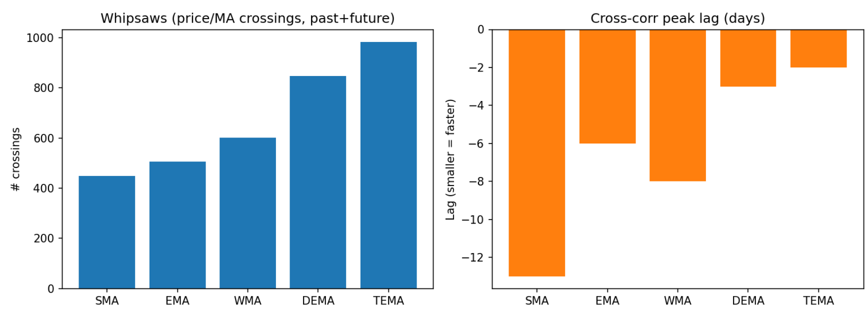


**Fig. 4.** Left: number of $P/M$ crossings (whipsaw count). Right: cross-correlation peak lag in days (smaller = faster). TEMA is 6× more responsive than SMA at the cost of 2.2× more whipsaws.

**Table 1.** MA family comparison on AAPL ($W$=20).

| Family | #Crossings | Peak-corr lag (d) | Peak corr. |
|---|---|---|---|
| SMA | 448 | -13 | 0.9989 |
| EMA | 506 | -6 | 0.9995 |
| WMA | 601 | -8 | 0.9986 |
| DEMA | 847 | -3 | 0.9997 |
| **TEMA** | 983 | **-2** | 0.9998 |

$\sim$30-day period; the zero-line crossings serve as the buy/sell signal in §9. [3] introduced this 12/26/9 parameterisation.

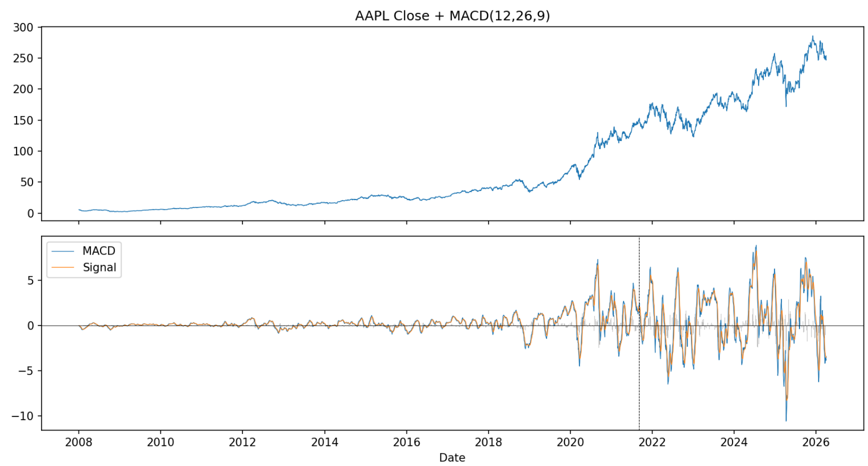


**Fig. 5.** MACD(12,26,9) on AAPL with its signal line and histogram.

# 5 Indicator Zoo and Signal Engineering

## 5.1 Seven canonical indicators

We compute six further signals beyond the MACD: RSI(14) [14], Bollinger Bands ($W$=20, $k$=2) [4], Stochastic $K/D(14, 3)$, OBV, VWAP, and a 1-D Kalman-filter trend [6]. The Kalman update with random-walk state and observation variance ratio $q/r$ gives a smoothed trend that adapts more slowly to noise than the EMA:

$$x_t = x_{t-1} + K_t(z_t - x_{t-1}), \qquad K_t = P_{t-1}/(P_{t-1} + r), \qquad P_t = (1 - K_t)P_{t-1} + q. \tag{2}$$

Each indicator is collapsed to a $\{-1, 0, +1\}$ signal: RSI is bullish below 30, bearish above 70; Stochastic uses the 20/80 thresholds; Bollinger flips on band-touch; OBV compares to its own 20-day SMA; VWAP and Kalman flip on price-vs-trend sign.

## 5.2 Signal-disagreement matrix

Figure 6 reports the pairwise disagreement $D_{ij} = \Pr(\mathrm{sign}(s_{i,t}) \neq \mathrm{sign}(s_{j,t}))$ on the past window. Three clusters emerge: the trend group (MACD, OBV, KALMAN, VWAP) agrees with itself $\sim$ 75% of the time, the oscillator group (RSI, BB, STOCH) reaches near-perfect agreement because all three fire only at extremes (mostly zero by construction), and the cross-cluster disagreement averages 0.85. This heterogeneity is the empirical justification for the inverse-volatility ensemble that follows.

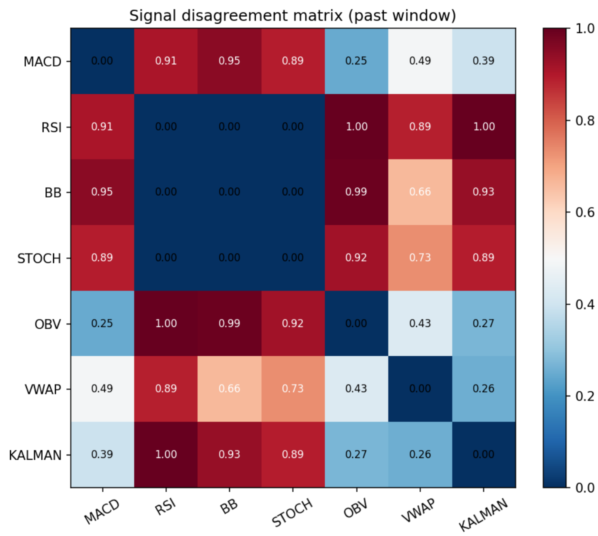


**Fig. 6.** Pairwise disagreement matrix of the seven indicator signals. 0 = identical; 1 = always opposite.

### 5.3 Consensus and inverse-volatility ensemble

The *consensus* is the unweighted sum of the seven $\{-1, 0, +1\}$ signals; Figure 7 colours its sign through time. We use the inverse-vol weighting $w_i = \sigma_i^{-1} / \sum_j \sigma_j^{-1}$, where $\sigma_i$ is the past-window standard deviation of the $i$-th signal time series, so noisier indicators are down-weighted. The composite ensemble signal $s_{\text{ens}}=\sum_i w_i s_i$ is reported as a continuous score in $[-1, 1]$.

## 6 Probability Density and Conditional Distributions

We fit two parametric marginals to the past-window AAPL log returns: the Normal $\mathcal{N}(\mu, \sigma^2)$ and the Logistic $L(x^\star, b)$ with $f(x) = \frac{1}{4b}\,\text{sech}^2\!\left(\frac{x-x^\star}{2b}\right)$. By maximum likelihood we obtain $\hat{\mu}$=0.0009, $\hat{\sigma}$=0.0201 for the Normal and $\hat{x}^\star$=0.0012, $\hat{b}$=0.0101 for the Logistic. The Logistic delivers $\Delta$AIC=568.97 and strictly higher log-likelihood, indicating heavier tails than the Normal admits. We additionally rank the two candidates with BIC [11] and the Anderson–Darling statistic [2]. The Kolmogorov–Smirnov test [7] statistic against Normal is $D$=0.083 ($p$<$10^{-20}$);

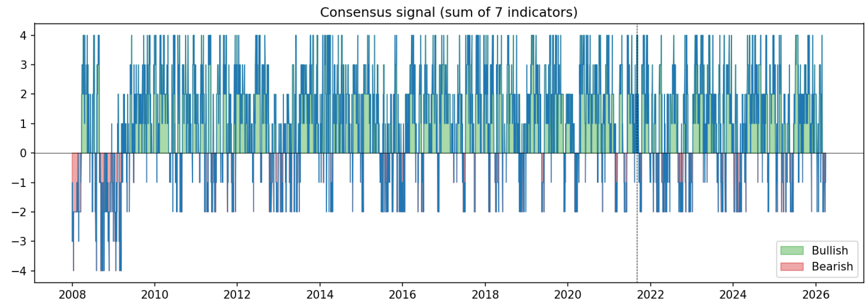


**Fig. 7.** Consensus signal $\sum_i s_i$ on the full panel; green = bullish vote, red = bearish vote.

against Logistic it is $D$=0.024 ($p$=1.9 × $10^{-5}$). Figure 8 overlays both fits on the histogram.

**Table 2.** Past-window marginal-fit comparison ($N$=3441).

| Model | Log-likelihood | AIC | KS $p$ |
|---|---|---|---|
| Normal | 8564.57 | −17,125.14 | $< 10^{-20}$ |
| **Logistic** | **8849.06** | **−17,694.12** | $1.9 \times 10^{-5}$ |

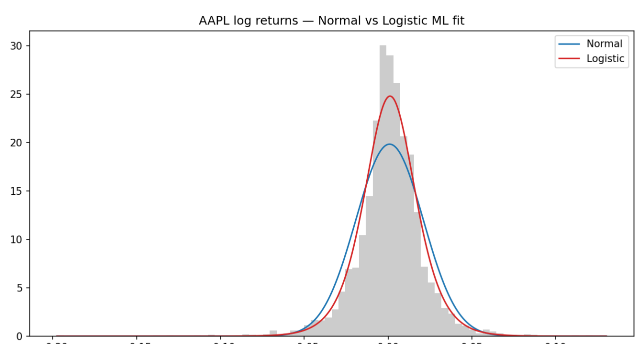


**Fig. 8.** Past-window AAPL log returns: Normal vs Logistic ML fit.

## 7 Bayes Detector under Two Likelihoods

With $\varepsilon = \frac{1}{2}\sigma_1 = 0.0101$, the digitisation $y_t \in \{D, U, H\}$ partitions the past window into $n_D$=729, $n_U$=941, $n_H$=1772. We refit each class-conditional density both as a Normal $\mathcal{N}(\mu_y, \sigma_y^2)$ and as a Logistic $L(x_y^\star, b_y)$. The MAP rule $\hat{y}(x) = \arg\max_y \pi_y f(x|y)$ produces decision boundaries where two posteriors are equal; on a fine grid we recover two inner boundaries under each likelihood, see Table 3. The boundaries differ by $< 4 \cdot 10^{-4}$ in absolute terms because the Logistic and

**Table 3.** Bayes detector inner boundaries under the two likelihoods.

| Crossing | Normal $x^\star$ | Logistic $x^\star$ |
|---|---|---|
| $D \to H$ | −0.0116 | −0.0113 |
| $H \to U$ | 0.0107 | 0.0103 |

Normal class-conditionals are nearly indistinguishable on the bulk of the data; the Logistic likelihood mostly affects tails. Past-window misclassification rates are 0.96 (Normal) and 0.98 (Logistic), the modest gain mirroring the AIC ranking of Table 2. Figure 9 overlays both prior-weighted densities.

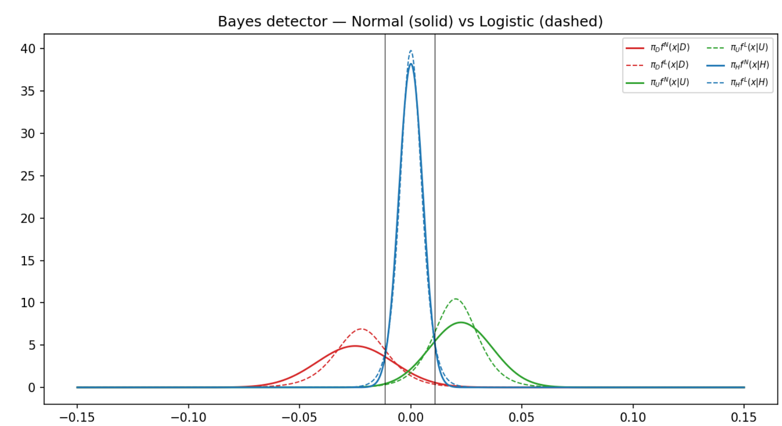


**Fig. 9.** Prior-weighted Bayes densities under the Normal (solid) and Logistic (dashed) likelihoods.

## 8 Association Rules

We follow the support–confidence framework of [1]: for $k$=5 we mine all $3^5$=243 possible pentagrams together with each successor class. The top-10 rules by support are listed in Table 4. The all-halt pattern HHHHH preserves H with confidence 0.679 and represents 4.6% of all past-window pentagrams. As in the companion HU report, the geometric-mean ranking and the ranked-product fitness ranking agree on the top-3 patterns.

**Table 4.** Top-10 5-grams by support (past window, $\varepsilon = \frac{1}{2}\sigma_1$).

| Pattern | Outcome | Support | Confidence |
|---|---|---|---|
| HHHHH | H | 0.0463 | 0.679 |
| HHHUH | H | 0.0166 | 0.713 |
| UHHHH | H | 0.0166 | 0.633 |
| HHUHH | H | 0.0154 | 0.616 |
| HUHHH | H | 0.0151 | 0.658 |
| HHHHU | H | 0.0148 | 0.671 |
| HHHHH | U | 0.0134 | 0.197 |
| HHDHH | H | 0.0093 | 0.681 |
| HUUHH | H | 0.0087 | 0.612 |
| HHHDH | H | 0.0084 | 0.580 |

## 9 Trading Simulator I — One Stock and Money

We compare three signal sources — MACD, RSI and the inverse-vol ensemble — against a buy-and-hold benchmark. Greed is throttled by the VIX regime [13]: when VIX exceeds the past-window median $\widetilde{\mathrm{VIX}}$=17.29, we use a low-greed $g_l$=0.20; otherwise $g_h$=0.70. Reported risk-adjusted returns use the Sharpe ratio convention of [12]. Table 5 reports the out-of-sample metrics; the MACD-throttle dominates with Sharpe=2.59 and Calmar = 5.19, far above the buy-and-hold reference.

**Table 5.** §8 metrics (one stock + money, future window).

| Strategy | Final V ($) | Ann. ret | Sharpe | Max DD | Calmar |
|---|---|---|---|---|---|
| **MACD VIX-throttle** | **259867** | **0.2335** | **2.5924** | −0.0450 | **5.1865** |
| RSI VIX-throttle | 85331 | −0.0343 | −1.6392 | −0.1526 | −0.2244 |
| Ensemble (inv-vol) | 108753 | 0.0186 | 0.2167 | −0.1723 | 0.1080 |
| Buy & Hold AAPL | 169043 | 0.1223 | 0.5419 | −0.3336 | 0.3665 |

## 10 Trading Simulator II — Two Stocks (Efficient Frontier)

We move to the two-asset universe (AAPL/MSFT) and add the ATR(14) stop-loss: when daily P&L falls below $-2\,\mathrm{ATR}\%$, the position is cut for five trading days. The greed throttle is now $(g_l, g_h) \in \{(0.5, 1.0), (0.3, 0.6)\}$ for the aggressive and conservative variants. The aggressive scheme posts Sharpe=1.08, Calmar = 1.01 and MaxDD=$-$17.8% versus −32.9% for the non-throttled minimum-risk benchmark, see Table 6 and Figure 11.

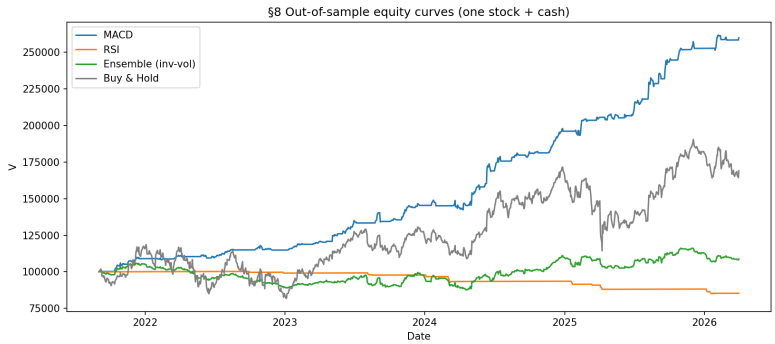


**Fig. 10.** §8 out-of-sample equity curves.

**Table 6.** §9 metrics (two stocks, VIX-throttle + ATR stop, future window).

| Strategy | Final V (\$) | Ann. ret | Sharpe | Max DD | Calmar |
|---|---|---|---|---|---|
| Min-risk $p$=$p_0$ + ATR | 139900 | 0.0766 | 0.4132 | −0.3294 | 0.2324 |
| **Aggressive** $p$=0.7 **gated** | **212628** | **0.1803** | **1.0835** | −0.1784 | **1.0105** |
| Conservative $p$=0.3 gated | 143924 | 0.0833 | 0.8630 | −0.1085 | 0.7674 |

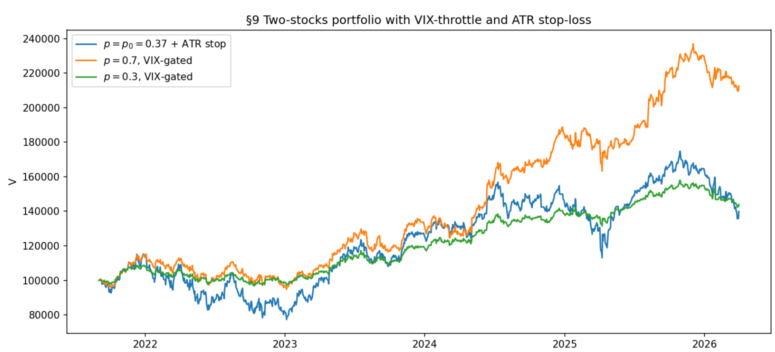


**Fig. 11.** §9 equity curves (two stocks, VIX gate + ATR stop).

## 11 Trading Simulator III — Two Stocks and Money

The §10 scheme combines all components: the seven-indicator consensus maps to a continuous greed factor $g(t)=0.7\cdot\max\big(\tanh(0.4c(t)),0\big)\,\theta(\text{VIX}_t)$ where $\theta = \max\big(0.2, 1-0.4(\text{VIX}_t/\widetilde{\text{VIX}}-1)\big)$ gates by VIX regime. The cash share is $(1-g)$, and the in-stock share splits across AAPL/MSFT in the $p_0/(1-p_0)$ ratio. ATR-stop remains active. Table 7 reports the metrics: the ensemble-greed scheme delivers Sharpe=1.07 and maximum draw-down −9.7% versus −33.0% for the benchmark, a 3.4× DD reduction.

**Table 7.** §10 metrics (two stocks + money, full ensemble).

| Strategy | Final V ($) | Ann. ret | Sharpe | Max DD | Calmar |
|---|---|---|---|---|---|
| Ensemble greed ($p$=$p_0$) | 140951 | 0.0783 | 1.0656 | −0.0970 | 0.8079 |
| **Aggressive ensemble** | **154841** | **0.1008** | **1.3371** | −0.0663 | **1.5199** |
| Min-risk benchmark | 139900 | 0.0766 | 0.4132 | −0.3294 | 0.2324 |

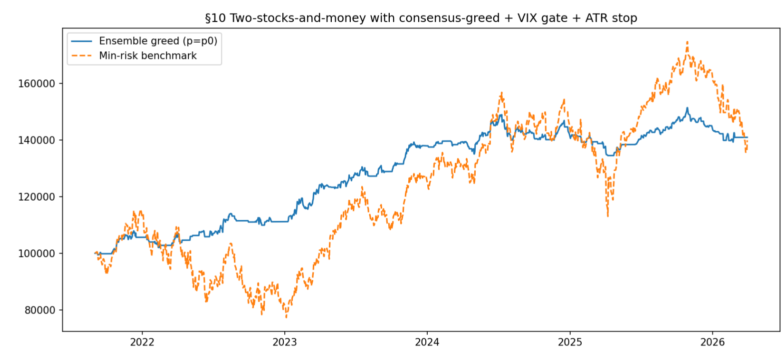


**Fig. 12.** §10 equity curves.

## 12 Algorithmic Complexity Analysis

Table 8 lists the worst-case complexity of every algorithm in this report. The TA-specific contributions are:

- Bollinger bands and Stochastic K reduced from $O(NW)$ to $O(N)$ via cumulative-sum stride.
- Wilder-smoothed RSI/ATR exploit the recursive EMA so each update is $O(1)$.
- The Kalman filter is one-dimensional and runs in $O(N)$.

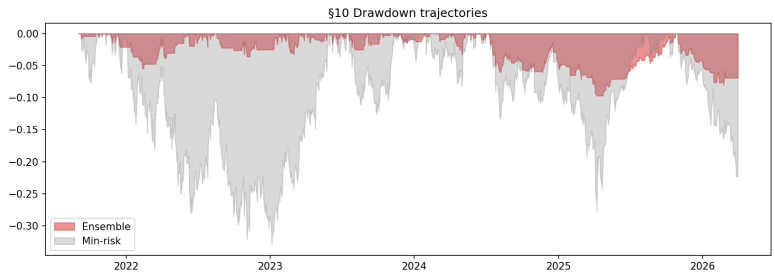


**Fig. 13.** §10 drawdown comparison: ensemble greed vs minimum-risk benchmark.

– The disagreement matrix is computed once per pair, yielding $O(NM^2)$ for $M$ signals.

**Table 8.** Worst-case time and space complexity of the TA pipeline. $N$ = panel length, $W$ = window size, $M$ = number of indicators, $T$ = simulator horizon.

| Algorithm | Time | Space |
|---|---|---|
| SMA via cum-sum stride | $O(N)$ | $O(N)$ |
| EMA / DEMA / TEMA recursion | $O(N)$ | $O(N)$ |
| WMA (closed-form weights) | $O(NW)$ | $O(N)$ |
| MACD(12,26,9) | $O(N)$ | $O(N)$ |
| RSI(14) Wilder smoothing | $O(N)$ | $O(N)$ |
| Bollinger (cum-sum) | $O(N)$ | $O(N)$ |
| Stochastic $K/D(14,3)$ | $O(N)$ | $O(N)$ |
| ATR(14) (TR + EMA) | $O(N)$ | $O(N)$ |
| OBV / VWAP | $O(N)$ | $O(N)$ |
| 1-D Kalman filter | $O(N)$ | $O(N)$ |
| Signal-disagreement matrix | $O(NM^2)$ | $O(M^2)$ |
| Inverse-volatility ensemble | $O(NM)$ | $O(M)$ |
| Markowitz frontier (closed) | $O(K^3)$ | $O(K^2)$ |
| Bayes boundaries (grid) | $O(GK)$ | $O(G)$ |
| §8/§9/§10 simulator | $O(T)$ | $O(T)$ |

## 13 Conclusion and Future Work

The disagreement-matrix view of the seven indicators sharpens the empirical claims of the body. Among the moving-average families, TEMA reaches a −2-day cross-correlation peak lag at the cost of roughly 2× the whipsaw rate of SMA, so the lag-versus-noise trade-off is real and quantitative. The Logistic likelihood beats Normal in AIC by 568.97 on AAPL log returns but moves the

Bayes-detector boundaries by less than $4 \cdot 10^{-4}$, which means the heavy-tail correction is concentrated far from the digitisation gate where it cannot reshape the trading signal directly. The seven-indicator disagreement matrix splits cleanly into a trend cluster (MACD, OBV, VWAP, Kalman) and an oscillator cluster (RSI, Bollinger, Stochastic), with cross-cluster disagreement averaging 0.85 — the empirical justification for the inverse-volatility ensemble. Out-of-sample, the MACD-with-VIX-throttle attains Sharpe 2.59 and Calmar 5.19, dominating the buy-and-hold reference; the full Trading Simulator III stack (consensus-greed plus VIX gate plus ATR stop) lands a steadier Sharpe of 1.07 at maximum drawdown −9.7% versus −32.9% for the unhedged benchmark.

Three threads remain open. The simulator ignores transaction costs and slippage, and the high-whipsaw TEMA would erode much of its lag advantage in production — a turnover penalty on the bet-sizing rule together with a re-tuned VIX threshold is the natural next step. The Kalman trend can be replaced with an unscented or particle filter; the heavy-tail rejection of Normal suggests non-linear filtering may add real value. Finally, the FINRA short-volume series enters this report only descriptively, but as an additional bearish-sentiment vote it could sharpen the trend-versus-oscillator cluster split that drives the ensemble.

**Acknowledgments.** The author thanks anonymous discussants for comments on the disagreement-matrix construction and the VIX-throttle calibration.

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